Does access to loans lead to differences in capital market investment decisions depending on the financial stand

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1 Introduction

Global wealth inequality has risen since the 1980s, and evidence points toward a rise in its concentration. For China, Europe, and the United States combined, the top 1% wealth share has increased from 28% in 1980 to 33% in 2019, while the bottom 75% share hovered around 10% (Zucman, 2019).

As the widening gap between rich and poor is a complex phenomenon, various hypotheses and perspectives on its causes can be found - ranging from globalization and technological progress to more specific explanations such as domestic policies.

Looking at the broad range of explanations, returns on capital and therefore investment decisions seem to act as crucial drivers, especially because returns on capital tend to outpace income growth, thus causing wealth concentration at the top (Piketty and Goldhammer, 2014). Fundamental to returns on capital is not only the question of access to the capital market but also the actual investment decision. Besides investing higher amounts in absolute terms rich people with unrestricted access to capital markets also tend to invest in riskier assets promising higher payouts, thus exacerbating inequality. However, research by Mishra et al. (2015) shows that inequality also leads to higher risk-taking by poorer sections of society. As such, the relationship between risk preferences and inequality is not yet fully understood.

Rising inequality carries a set of negative consequences including the potential to increase violence and conflicts, causing financial crises and hindering economic development. One instrument that can help alleviate inequality are capital markets since they allow people to ease their budget constraints through borrowing and investments. For example, during the COVID - 19 crisis, government handouts were provided that led to an increase in the participation in capital markets (Fitzgerald, 2020). Recent research by Rashid and Intartaglia (2017) has suggested that financial development, including saving mobilization and capital allocation has a positive impact on poverty reduction.

However, capital markets are usually far from complete and impose borrowing constraints on the poor. One argument against the relaxation of these constraints would be that easier access to loans in particular might cause them to be used in an inefficient manner or overused for non productive purposes.

Our paper aims to look at such issues, where we try to address the linkage between the access of capital market and the difference in investment between rich and poor people. We try to measure how richer and poorer people make decisions by allowing them to take loans. We manipulate this process by endowing different groups with differing amounts of money, with richer people having a higher endowment as compared to poorer people. This leads to our research question:

Does access to loans lead to differences in capital market investment decisions depending on the financial stand?

While there are many different possible approaches to investigate said question, our paper will largely build on Shah et al. (2012) whose research highlights the fact that attention and budget constraints play a role in explaining the behavior of economically disadvantaged agents. We expand on their work by considering risk preferences in decision-making while applying a real-life context in running a capital market focused experiment. We will build our hypotheses mainly on evidence from Shah et al. (2012) which implies that the relaxation of the budget constraint (through access to a capital market) affects economic groups differently. While richer individuals do not change their investment decisions significantly when receiving access to affordable loans, poor individuals might engage in risk-seeking behavior in the form of over-borrowing as they attempt to insure against financial shocks.

The paper is structured in the following way: First, we present the previous research and literature that may be relevant to understanding our approach. Second, we explain the design and structure of the experiment itself. Third, we have the discussion section in which we elaborate on the different hypotheses, power analysis as well as implications and potential shortfalls of our paper. Finally, we have a short conclusion of our paper.

2 Literature Review

So far, there have been different attempts in empirical research trying to explain how wealth and economic inequality can affect economic decision making. Past research illuminates various aspects of this relationship, as can be seen in a more detailed review of the literature that follows below.

Economic inequality and risk preferences

Previous studies have found evidence of economic inequality inducing risk taking behavior. Mishra et al. (2015) examined how individual risk taking behavior changed with economic inequality by conducting two experiments where the participants were exposed to different levels of economic inequality and then tasked to chose between options with varying levels of risk and reward. The inequality was simulated by allocating different initial endowments to participants, creating disparities in their initial resources. The study found that those individuals that were victims of inequality tended to be more inclined to take risk than the beneficiaries of inequality or those who were not exposed to economic inequality at all. Payne et al. (2017) found similar evidence. The researchers used 811 participants to conduct three experiments to examine this relationship, and in addition to this they used large scale internet data to analyze the risk behavior in relation to income inequality. The results of both the experiments and the analysis of the internet search data indicated that risk-taking behavior increased with more income inequality. The researchers suggests that this behavior is due to an upward social comparison which increases perceived needs.

Risk aversion and wealth

Besides the research showing how risk taking increases with inequality, the literature on risk preferences and their relation to wealth has been growing for the last couple of decades.

For example, Carroll (2000) examined the financial portfolio of households in the U.S. and found that affluent households tend to hold riskier assets in relation to the median households portfolio. The compositions of the wealthier individuals' portfolios differed in other ways as well: Affluent individuals are more likely to self finance their own entrepreneurial ventures where much of their net worth is laid in. However, the author argues that these findings do not necessarily support the idea of a negative relation between risk aversion and wealth. He rather argued that capital market imperfections and the assumption of bequests being luxury goods better explains the results from the study. Guiso and Paiella (2008) examined risk aversion in connection to wealth and exposure to background risk. They tested how much an individual was willing to pay for security against risk by using data from 1995 covering Italian households. The findings suggested a negative relationship between risk aversion and wealth, though the elasticity of risk aversion was estimated to be less than one, suggesting that the decrease in risk aversion is itself reducing with wealth. An older field experiment by Binswanger (1980) studied attitude toward risk in 240 households in rural India. Binswanger used two different methods; an interviewing approach and an experiment with a gambling game. Although the interview method suffered from bias and inconsistencies, the results from the experiment showcased that moderate risk aversion existed in all individuals when the reward was high. The risk aversion decreased with wealth, however this effect was not statistically significant.

Poverty and economic behavior

There is also a multitude of research on how the state of poverty can affect economic behavior. Haushofer and Fehr (2014) reviews a body of literature supporting the hypothesis that poverty affects cognitive function and economic decision-making, making it difficult to escape poverty. The evidence examined is consistent and points in the same direction: Poverty, particularly the factor of scarcity, largely causes stress, difficulty with attention, and may lead to short-sighted and risk-averse decision-making.

In the paper by Shah et al. (2012), the authors aim to explore how scarcity can influence economic behavior, such as focusing primarily on immediate issues while neglecting others. The researchers argue that this results in irrational economic decision-making, such as over-borrowing, which, in turn, reinforces poverty. The paper thus attempts to show that scarcity creates a tendency to focus on pressing expenses in the moment and, consequently, a tendency to borrow without fully assessing whether the benefits outweigh the costs. The researchers demonstrate this by conducting several experiments that showcase how scarcity alters attention span and causes a shift in focus toward specific problems. The experimental design of this paper is built on a series of different lab experiments testing the effect of scarcity, where some participants are exposed to constraints that simulate scarcity, while others are not. A weakness of this study is the arguably small sample sizes in some of the experiments.

The researchers also performed a replication of the experiments in Shah et al. (2019), where they were able to confirm the findings related to attention focus and over-borrowing under scarcity. However, results from the replication regarding the hypothesis that scarcity leads to attentional neglect were not significant, and they found no evidence of cognitive fatigue, as reported in the original paper.

By specifically examining investment decisions in capital markets, we aim to investigate how the aspects discussed above might influence individuals' investment behavior. Building on a similar experimental design approach as Shah et al. (2012), our paper will contribute to the existing body of literature on wealth, inequality, and poverty, as well as their effects on economic decision-making, while applying a context that is more relevant to real-life financial situations.

3 Experiment

Motivated by these diverging angles and hypotheses aimed at explaining inequality and differing investment behaviors within socioeconomic groups, we contribute to the existing research with an investment game testing how investment decisions and borrowing behavior interact with financial standing.

Similar to the experiment of Shah et al. (2012) we will fist randomly assign participants to one out of four groups with two different endowments modeling exposure to scarcity in the financial context (poor / rich) and access to loans (no access / access). Subsequently, we depart from the authors' game-based approach: Participants receive an endowment each round (income) and can decide to invest in risky assets to increase their wealth, which is carried from round to round. Apart from thus engaging participants in real-life financial decision-making, we expose all participants to the risk of negative income shocks, aiming to investigate the real-life implications that Shah et al. (2012) allude to.

Following the implication from Shah et al. (2012) and their own replication study (Shah et al., 2019), we expect poor individuals to exhibit excessive borrowing when having access to loans, owing to resource scarcity and stronger relative effects of budget constraint-relaxation on poorer as compared to richer participants. Additionally, in line with empirical research, we expect richer individuals to invest a comparably larger share of their disposable income in the riskier asset, irrespective of their access to loans.

3.1 Setup

In 2020 the US government allocated payment checks[1] to their citizens generating a budget relaxation during the COVID-19 pandemic. This policy significantly increased capital market participation across socioeconomic groups (Fitzgerald, 2020). Referencing to these quantifiable outcomes, our experiment will play in a context similar to that faced by Americans.

In line with the replication study from Shah et al. (2019) we will hire participants from

[1]Coronavirus Aid, Relief, and Economic Security (CARES) Act

MTurk with their locality based in the United States. The recruited subjects will receive a participation endowment of at least \$5 with an additional payout contingent on performance in game (final wealth, see Section 3.3)[2]. As for the number of participants to be recruited, see our power analysis in Section 4.2.

Each participant will be randomly assigned to one of four groups:

Table 1: Participant groups differentiated by income and loan access

Group	Disposable income (each round)	Access to loans (each round)
1	\$250 (poor)	–
2	\$250 (poor)	\$0 to \$125 (5% interest)
3	\$500 (rich)	–
4	\$500 (rich)	\$0 to \$250 (5% interest)

Participants do not learn of their group assignment prior to the commencement of the game. Joining the experiment, they will immediately be exposed to the particulars of the game, as detailed in Section 3.2.

3.2 Game

Prior to the game's commencement, participants are provided with an overview of the game. They receive information on:

- Number of rounds to be played (10)
- Endowment that they, individually, receive each round (income)
- Access to and interest due for loans, if applicable
- Investment decision they will face each round (save or invest in stock)
- Risk of negative income shock (breakdown of personal car required for commute)
- Indication on payouts (\$5 min, \$10 for 0 wealth at game-end, \$20 max)

Looking at the literature we will abstain from telling the participants whether they are poor or rich in order to avoid effects influencing their behavior: poor people feel left behind

[2]Linking the investment games' payouts to the real-live insures that participants will make representative decisions.

which can lead to irrational behavior displayed by extreme risk-seeking, as discussed in Section 2.

The game proper will extend over 10 rounds, each round adheres to a uniform pattern [3]. At the beginning of each round, participants receive their income as per their group assignment (see Table 1). Participants are displayed their current in-game wealth measured in $ at every point throughout the game.

Prior to taking any decisions, participants are reminded of the income-shock they may face at the end of the round. With a 20% chance, their personal vehicle may break down ensuing $400 for towing and repairs.[4] Repairs cannot be delayed, as the car is required for their commute to work. Car troubles were selected for framing the income shock, as they constitute the most pertinent day-to-day feature in the US. Other income shock examples such as medical bills are prone to being put off, household appliances breaking down might be temporarily substituted.

Subjects assigned to having access to loans are offered to take out a loan of up to 50% of their income, as per Table 1. The choice of loan amount is in $5 steps and undertaken via a slider. Participants are informed that loans must be repaid prior to the following round. The exact repayment amount of principal and 3% interest is specified in line with the loan choice on the slider (e.g. if a loan of $100 is selected, participants are informed that repayment of $103 is due prior to the next round).[5] If a loan is taken out, the corresponding amount is added to a participant's displayed wealth.

Participants are subsequently faced with an investment decision. They are free to invest 0% to 100% of their wealth into a risky asset (stock). Asset specifications are displayed to participants as in Table 2. The expected return on investment is therefore 9%, of which participants are not informed. The expected return therefore outstrips the bank loan interest, but is associated with a considerable downside risk.

[3] In order to avoid bias by fatigue effects caused by different amounts of rounds played as remarked by Shah et al. (2019).

[4] This magnitude alludes to findings pointing out that 37% of Americans are not able to pay for a $400 income shock (Board of Governors of the Federal Reserve System, 2023).

[5] We opt for a low-interest financial instrument to be able to investigate a genuine budget-constraint expansion, similar to the aforementioned US stimulus checks of the Covid-19 pandemic period.

Table 2: Participant information on investment-decision in stock

Probability	Outcome	Example: $100 invested
60 %	+25%	$125
30 %	−10%	$90
10 %	−30%	$70

In accordance with Shah et al. (2012), and the confirmation of results in this respect in Shah et al. (2019), the time needed to place financial decisions on loan-taking and investment is measured separately[6]. Hypotheses in this regard are discussed in Section 4.

Following the taking of the investment decision, the round concludes. Upon each conclusion of a round of the game, participants are informed of the respective realizations in $-amounts of:

- income shock (±$0/−$400),
- investment (if any: +25%, −10%, or −30% of their investment), and
- loan repayment (if any: principal and 3% interest)
- their resulting net wealth

Participant's net wealth is subsequently carried over to the next round and continues to be displayed. A negative net wealth (in disposable income!) is possible, in line with 8.4% of American households having negative net worth (). Negative wealth is not subject to an interest mirroring the widespread use of no-interest credit cards to continuously meet monthly expenses. Should a participant hold negative wealth at the beginning of a round even after receiving their endowment, they can only take out a loan to undertake an investment decision (if they have access to the loan instrument). Each participant thereby plays the same number of rounds in order to preclude any effects from cognitive burden, as discussed by Shah et al. (2019) in reviewing their previous work.

Upon conclusion of the 10$^{\text{th}}$ round of the game, participants are informed of their resulting net wealth and the corresponding payout they will receive.

[6]In line with Shah et al. (2012) time taken to decide on the investment should increase, since resource scarcity should amplify engagement.

3.3 Payouts

The baseline for the payout of each player is \$10, the amount received in case a net wealth of \$0 is held at end of round 10 of the game. Players are informed that their payout may go as low as \$5 if their resulting net wealth is negative, and reach up to \$20 for high positive net wealth. The payout schedule is not specified further for participants.[7]

Departing from Shah et al. (2012), we "punish" negative wealth realizations to ensure incentive compatibility at all wealth levels, i.e. prevent reckless behavior by participants who find themselves indebted. Equally, an increasing payout that is not capped by an in-game \$-amount motivates sensible handling of resources even for participants in the "rich" group. The importance of such incentive compatibility is highlighted by an extensive body of literature, e.g. (Cox and Sadiraj, 2018).

[7] See the exact schedule mapping final participant wealth to payouts in Section A.

4 Discussion

The following section will provide a discussion of the implementation and the most likely outcomes of our experiment. Throughout this section, we will refer to the high endowment group as "rich" and the low endowment group as "poor". Similarly, access to the capital market (synonymous with treatment) means having the option of taking on a loan, irrespective of whether the individual ends up taking it or not. Furthermore, all indications of $ relate to real-life currency (as opposed to in-game currency).

4.1 Hypotheses

Looking at previous literature we expect the following hypotheses to hold:

- (i) rich participants tend to invest more of their money (as a share of their endowment) in assets than poor subjects, regardless of their access to loans
- (ii) borrowing of poor individuals with access to the capital market is irrationally high, thus leading to lower final payouts (over-borrowing)
- (iii) negative income shocks focus minds, but more so when resources are scarce - i.e. participants should take longer to make a decision in the subsequent round.

Hypothesis (i) references directly to empirical analyses in existing literature whose results suggest that we should find a highly significant effect. This would be in line with the findings of Carroll (2000), where a positive relationship between higher financial stand and riskier assets held in the US-population was found. Another argument supporting hypothesis (i) to hold would be the results of Guiso and Paiella (2008), providing evidence for a negative correlation between wealth and risk aversion.
In statistical terms, we can run a regression of the following structure

$$a_{i,t} = \alpha_0 + \alpha_1 R_i + \alpha_2 T_i + \alpha_3 R_i T_i + u_{i,t}$$

where a is the proportion of the disposable income that individual i has invested in assets in round t, R is a dummy variable for individual i being rich, T_i indicates their treatment status (access to the capital market) and $u_{i,t}$ is an i.i.d. error term. Hypothesis (i) would then be stated as $H_0: \ \alpha_1 \leq 0$ vs. $H_1: \ \alpha_1 > 0$ and tested accordingly. To analyse whether access to the capital market influences this behavior, we also include an interaction term to capture such an effect (which we expect to be small or close to zero).

Hypothesis (ii) alludes in particular to the study of Shah et al. (2012) concluding that poorer individuals tend to borrow more and engage in over-borrowing as they face a scarcity of resources. This scarcity causes cognitive stress and an increased mental load, resulting in participants directing their focus towards more salient problems irrespective of their rational priority. The misallocation of attention leads to a neglect of less salient yet relevant issues, the consequences of which can be detrimental especially in the long run. As a result, we expect poor individuals (suffering from resource scarcity) to make more irrational choices - thus exhibiting a greater overall variation in their behavior - and specifically to engage in over-borrowing as was found in the study of Shah et al. (2012). As such, access to the capital market (the treatment T_i should have a negative impact on final payouts that is especially pronounced for poor participants. This effect can be assumed to be of significant size since it was later confirmed in the replication study by Shah et al. (2019). In order to estimate the difference in final payouts, we can and use the following difference-in-differences model:

$$Y_i^{final} = \beta_0 + \beta_1 R_i + \beta_2 T_i + \beta_3 R_i T_i + v_{i,t}$$

Here, Y is the final payout that individual i receives after the last period measured in Euros (ranging from 5 to 20), R_i once again indicates whether individual i is rich, T indicates their treatment status (1 if access to capital market) and $v_{i,t}$ is an i.i.d. error term. Since we expect access to the capital market to be especially detrimental to the poor individuals' performance, we can formulate our hypothesis (ii) as $H_0: \ \beta_3 \geq 0$ vs. $H_1: \ \beta_3 < 0$ and run the corresponding tests.

Thirdly, Shah et al. (2012) also finds that poor individuals tend to take more time for their actions than rich people do. If this applies to our experiment as well, we should find that more time elapses between rounds if fewer resources are available. The reason for this behavior can once again be found in the increased cognitive stress under resource scarcity: If fewer resources are available, mental load increases and the completion of tasks takes more time than usual. For our statistical analysis, we need to measure the time S in seconds during the decision stage of each round except the first and test whether the decisions of the participants take longer if there was a negative income shock at the end of the preceding round. We also wish to understand whether this effect is particularly strong for poor individuals, which is why we introduce an interaction term leading to a difference-in-differences setup of the following structure

$$S_{i,t} = \gamma_0 + \gamma_1 N_{i,t-1} + \gamma_2 R_i + \gamma_3 N_{i,t-1} R_i + w_{i,t}$$

where N is a dummy variable for a negative income shock affecting individual i in round t and w is an i.i.d error term. Following the intuition that a negative income shock in the previous round causes greater resource scarcity and thus longer decision times especially for the poor, hypothesis (iii) can be restated as $H_0: \ \gamma_3 \leq 0$ vs. $H_1: \ \gamma_3 > 0$.

4.2 Power Analysis

To ensure that our experimental setup is capable of delivering meaningful results in terms of statistical significance, we need to consider the importance of choosing the right sample size. However, the question of sample size depends on several factors that are not easy to determine a priori. For example, the sample size needed to achieve a power of 80% at a significance level of 95% with would be

$$n = \frac{(Z_{0.8} + Z_{0.95})^2 \cdot \sigma^2}{\beta_1^2 \cdot p(1-p)}$$

where σ^2 is the variance of the outcome variable, β_1 is the assumed effect size and p is the proportion of the groups we are comparing. Regarding the latter we can reasonably argue that the sample size of rich and poor as well as treated and untreated individuals should be equal ($p = 0.5$) in order to ensure that the degree of randomization can be assumed identical in all groups.

In the case of hypothesis (i), we have an outcome variable that is bounded ($a_{i,t} \in [0,1]$), therefore we can choose the maximum possible variance (0.25) for σ_α^2 to err on the safe side. If we now expect rich individuals to invest 20 percentage points more of their income than poor individuals, the sample size needed to detect such effect in 80% of cases at a significance level of 95% would therefore be

$$n = \frac{(0.84 + 1.96)^2 \cdot 0.25}{0.1^2 \cdot 0.5(1 - 0.5)} = 196$$

In testing hypothesis (ii), we again have a bounded outcome variable ($b_{i,t} \in [5, 20]$) which allows us to choose the maximum possible variance of $(20-5)^2/4 = 56.25$ for σ_β^2. However, we should also consider that the coefficient we're looking for only applies to one quarter of the total sample (those that are poor and treated), which is why we need to set $p = 0.25$. Assuming that we expect access to the capital market to lower the average final payout of the poor by \$3 (in real-life currency) more than it would for the rich counterparts, the sample size needed to detect such effect in 80% of cases at a significance level of 95% would be

$$n = \frac{(0.84 + 1.96)^2 \cdot 56.25}{3^2 \cdot 0.25(1 - 0.25)} = 262$$

For hypothesis (iii), we can follow the same logic as previously with the exception of now having a subgroup ratio of $p = 0.1$. This is because the relevant coefficient only applies to individuals that are hit by a negative income shock (20% probability) while being poor (50% probability). If we set an upper limit of 20 seconds for the investment decision and expect the income shock to prolong the average decision time by 5 seconds more for the poor compared to the rich, the sample size needed to detect such effect in 80% of cases at a significance level of 95% would be

$$n = \frac{(0.84 + 1.96)^2 \cdot 100}{5^2 \cdot 0.1(1 - 0.1)} = 349$$

.

We can thus conclude that a reasonable minimum sample size would be around 400 participants in total in order for the tests to produce meaningful results. Any number above that would of course allow for a more reliable detection of smaller effects than those stated in the calculations. However, we also need to consider the risk of different baseline risk preferences in each of the groups (rich vs. poor, treatment vs. no treatment). For the sake of simplicity, we can look at the share of above-median risk averse people (which, by definition, should be 50%). In a subsample size of 200 per group (to compare rich vs. poor for example), the probability that the share of above-median risk averse averse people differs by more than 10% between the two subsamples is less than 5%:

$$z = \frac{0.1}{\sqrt{0.25 \cdot \frac{1}{400}}} = 2 \quad \rightarrow p(|z| > 2) = 0.0456$$

Although this is a very simplistic calculation, we can assume that the risk of having different baseline risk preferences in our subsamples is manageable at a sample size of 400. Another way of dealing with this issue would involve eliciting the individual risk aversion and balancing the subsamples accordingly.

The cost of running the experiment depends on the payouts of the participants which are in turn randomly affected by negative income shocks, asset payoffs and loan taking. Considering the minimum and maximum remuneration of \$5 and \$20, respectively, this creates a total cost window of \$2000 to \$8000 at a sample size of 400. However, conversion rates from the in-game currency to real-life currency could be somewhat adjusted in order to manage the expected costs.

4.3 Implications of the findings

If we look at hypothesis (i), a rejection of the null hypothesis would confirm our assumption that wealth is correlated with lower risk aversion, as was shown in previous literature (see Carroll 2000, Guiso and Paiella 2008). Should we fail to reject the null hypothesis, our results could imply that it is not merely the endowment effect of having more resources that drives the change in risk behavior and that it is instead something that lies in the nature of (real-life) rich individuals. In consequence, this would give reason to assume that it is not the wealthiness that makes individuals more risk-loving but rather the risk affinity that leads to their accumulation of wealth. On the other hand, it could also be that differences in financial literacy are explanatory factors to the behavior we observe in empirical data. Lastly, it will also be interesting to see whether capital market access changes the investment behavior or rich individuals and how this change compares to that of the poor individuals. We expect the corresponding coefficient to be relatively small since rich individuals should not adapt their investment behavior to their loan access treatment as much as poor individuals might do.

Regarding hypothesis (ii), rejecting the null would particularly support the previous findings of Shah et al. (2012) as they found poor participants to engage in over-borrowing leading to a worse performance than without the option of borrowing resources. In the context of our experiment, we would then find that poor actually become poorer while the rich potentially become richer. In that case, our results would be able to shed some light onto the ever increasing wealth inequality we see globally and the mechanisms behind it. If, however, the null hypothesis cannot be rejected, this would subsequently cast some doubt on the findings by Shah et al. (2012), although one should consider that their experiment took place in a somewhat different context than ours - hence providing some explanation as to why their results could also differ from ours.

A rejection of the null in hypothesis (iii) would once again confirm the findings of Shah et al. (2012) and reaffirm the intuition of resource scarcity increasing mental load, thus prolonging decision times. Any different finding, i.e., a failure to reject the null hypothesis, would indicate that poor individuals suffering from a negative income shock spend less time making their decisions. A possible explanation could be individuals with a negative net wealth that are unable to invest in the asset according to the rules and thus have no decision to make (if they are unwilling or unable to take a loan).

4.4 Potential shortfalls of the experiment

As with any experimental study, there are a few potential threats to identification that need to be addressed.

First and foremost, any lab experiment inevitably calls for the question of external validity. Compared to Shah et al. (2012) and Shah et al. (2019), we have adapted the narrative of our study to better fit the real-life financial circumstances that individuals may face, thus making our results somewhat more applicable. However, our review of the previous literature has also revealed that the relationships in question are far from unambiguous and reveal a level of complexity that may not be obvious from the onset. As such, we cannot expect all of our findings to translate directly to empirical data and further research would be needed to investigate this issue.

In addition, there are also some concerns of more technical nature. For one, the risk of different baseline risk preferences between the subsamples could introduce significant bias to the results. If, for example, the rich group has a lower baseline average risk aversion than the poor group, it might cause the rich individuals to invest in the asset for reasons that have nothing to do with being rich in the sense of our experiment. While this risk is somewhat manageable as shown in section 4.2, it is not easy to deal with and nearly impossible to avoid entirely. The only reliable countermeasure would be a very large sample size to ensure proper randomization, however, this could be problematic from a practical and financial perspective.

Another potential issue is that the figures regarding endowments, payouts and loan costs might not be well enough calibrated. If, for example, poor individuals would never invest in the asset simply because they lack the means to do so, this would make it difficult to attribute changes in their final payouts to their borrowing and investment behavior.

Furthermore, from a statistical perspective, there is a lot of random variation in the outcome variables caused by, e.g., negative income shocks, asset payoffs, random behavior of individuals, and the like. While these sources of variation could be dealt with using more sophisticated models, the concern remains that it becomes more difficult to discern significant effects and random variation, especially in small samples.

Lastly, even though we tried to minimize the complexity of the game, we cannot completely rule out that some of the participants may not fully understand all aspects of the game and their implication - i.e., there is some minimum financial literacy required to understand the optimal choices within the game.

5 Conclusion

With rising inequality all over the world concerns about leaving the poor behind becomes more relevant than ever. This increase in disparity is not only precarious from an ethical but also from an economic perspective one. In order to alleviate and prevent the downsides of this trend there has to be a shift of attention towards the causes of this phenomenon.

With literature providing ample research about different potential factors driving the increase in inequality, there seems to be little consensus as to what could be the root cause of the issue. Therefore, we conducted an experiment aiming to gauge the significance of the most plausible hypothesis, negative wealth effects of resource scarcity and the correlation between risk preferences and investment decisions.

In accordance with the findings in previous literature (Shah et al. 2012 and 2019, as well as Carroll 2000 in particular), we have developed three hypotheses that we expect to hold. More specifically, we expect rich individuals to invest in riskier assets, poor individuals to engage in over-borrowing when given access to loans and resource scarcity to increase the focus on investment decisions. By investigating these hypotheses, we aim to shed more light on the factors that influence investment behavior under varying levels of resource constraints.

Apart from a detailed outline of the experimental setup, we have also provided guidance on the practical implementation of our study by conducting a power analysis as well as an estimation of the expected costs. However, there are still some potential shortfalls of our experiment that have been addressed as well. While the setup of our study certainly provides a more realistic and applied context than that of Shah et al. (2012), it must be kept in mind that any lab experiment can only cover a fraction of the true mechanisms at work. As such, our study calls for more research in the future that could perhaps investigate real-life income shocks (such as US income checks) in field studies to help understand how our findings might relate to empirical data.

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A Mapping of the payout

The investment in the asset can realize three different outcomes with an expected value of $109. In order to provide incentives high enough to ensure realistic behavior in the

Table 3: Possible outcomes of the assets

Probability	Outcome	Expected Value
0.6	125	109.0
0.3	90	
0.1	70	

experiment as well as staying within a reasonable budget we mapped the payouts to the earnings in the investment games as follows:

In-game wealth	Payouts
-$500 and below	5
-450	5.5
-400	6
-350	6.5
-300	7
-250	7.5
-200	8
-150	8.5
-100	9
-50	9.5
0	10
450	10.5
900	11
1350	11.5
1800	12
2250	12.5
2700	13

In-game wealth	Payouts
3150	13.5
3600	14
4050	14.5
4500	15
4950	15.5
5400	16
5850	16.5
6300	17
6750	17.5
7200	18
7650	18.5
8100	19
8550	19.5
$9,000 and above	20

With the minimum payout being the $5 participation payment and $20 being the maximum payout.